

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Bachelor of Technology (Mechanical Engineering) SEMESTER - 6 Summer 2025 ( Regular )**

**Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology**

**Semester : SEMESTER - 6**

**Subject Code & Name: BTMC602 - MACHINE DESIGN-II**

**Time : 3 Hours]**

**[Total Marks : 60**

**Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

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|                                                                         |                               |   |                                       |
|-------------------------------------------------------------------------|-------------------------------|---|---------------------------------------|
| 1 For a bearing subject to pure radial load, equivalent dynamic load is |                               |   |                                       |
| A                                                                       | $P = X \cdot Fr + Y \cdot Fa$ | C | $P = Fa$                              |
| B                                                                       | $P = Fr$                      | D | $P = X \cdot V \cdot Fr + Y \cdot Fa$ |

|                                                                                |                                |   |                                              |
|--------------------------------------------------------------------------------|--------------------------------|---|----------------------------------------------|
| If both pinion and gear are made of same material, then, in the design process |                                |   |                                              |
| A                                                                              | pinion is considered as weaker | C | both pinion and gear is considered as weaker |
| B                                                                              | gear is considered as weaker   | D | we cannot decide                             |

|                                                                                           |            |   |                   |
|-------------------------------------------------------------------------------------------|------------|---|-------------------|
| 3 Which type of gears are used to transmit power between two shafts having parallel axis? |            |   |                   |
| A                                                                                         | worm gear  | C | Spur gear         |
| B                                                                                         | bevel gear | D | None of the above |

|                                                                      |                               |   |                                       |
|----------------------------------------------------------------------|-------------------------------|---|---------------------------------------|
| For a bearing subject to pure axial load, equivalent dynamic load is |                               |   |                                       |
| A                                                                    | $P = X \cdot Fr + Y \cdot Fa$ | C | $P = Fa$                              |
| B                                                                    | $P = Fr$                      | D | $P = X \cdot V \cdot Fr + Y \cdot Fa$ |

|                                                                                                                                           |                               |   |                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|---|----------------------------------------|
| As per Lewis equation for beam strength, if $m$ and $b$ are same for pinion and gear, then weaker element will be decided on the basis of |                               |   |                                        |
| A                                                                                                                                         | Least magnitude of $\sigma_b$ | C | Least magnitude of $\sigma_b \times Y$ |
| B                                                                                                                                         | Least magnitude of $Y$        | D | Least magnitude of $\sigma_b / Y$      |

|   |                              |                                                         |   |                                                                                       |
|---|------------------------------|---------------------------------------------------------|---|---------------------------------------------------------------------------------------|
| 6 | Module of a gear is given as |                                                         |   |                                                                                       |
|   | A                            | Inverse of diametral pitch                              | C | Product of diametral pitch and circular pitch                                         |
|   | B                            | Ratio of no. of teeth (Z) to pitch circle diameter (d') | D | distance measured along the pitch circle between two similar points on adjacent teeth |

|   |                                                                                  |   |                         |  |
|---|----------------------------------------------------------------------------------|---|-------------------------|--|
| 7 | Which of the following is the expression for centre-to-centre distance in gears? |   |                         |  |
| A | $(d_p \times d_g) / 2$                                                           | C | $m(Z_p + Z_g) / 2$      |  |
| B | $(d_p - d_g) / 2$                                                                | D | $m(Z_p \times Z_g) / 2$ |  |

|                                                                                         |            |   |                   |
|-----------------------------------------------------------------------------------------|------------|---|-------------------|
| Which type of gears are used to transmit power between two shafts having parallel axis? |            |   |                   |
| A                                                                                       | worm gear  | C | Helical gear      |
| B                                                                                       | bevel gear | D | None of the above |

|   |                                                   |   |                                  |  |
|---|---------------------------------------------------|---|----------------------------------|--|
| 9 | The effective load on spur gear tooth is given by |   |                                  |  |
| A | $P_{eff} = C_s \times P_t \times C_v$             | C | $P_{eff} = C_s \times P_t / C_v$ |  |
| B | $P_{eff} = C_s / P_t \times C_v$                  | D | $P_{eff} = C_s \times P_t - P_d$ |  |

|    |                                                                              |   |                     |  |
|----|------------------------------------------------------------------------------|---|---------------------|--|
| 10 | Minimum number of teeth required in a gear to avoid interference is given as |   |                     |  |
| A  | $2 \times \sin^2 \alpha$                                                     | C | $2 / \sin 2\alpha$  |  |
| B  | $2 / (\sin \alpha)^2$                                                        | D | $2 / \sin^2 \alpha$ |  |

|                                     |               |   |                      |
|-------------------------------------|---------------|---|----------------------|
| Gear ratio can also be expressed as |               |   |                      |
| A                                   | $i = n_p/n_g$ | C | $i = Z_p / Z_g$      |
| B                                   | $i = n_g/n_p$ | D | $i = Z_p \times Z_g$ |

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|                                                                                                           |    |   |    |
|-----------------------------------------------------------------------------------------------------------|----|---|----|
| If pressure angle is $14.5^\circ$ in full depth involute system, then minimum number of teeth required is |    |   |    |
| A                                                                                                         | 18 | C | 17 |
| B                                                                                                         | 32 | D | 11 |

Q2. Solve the following.

A single-row deep groove ball bearing has a dynamic load capacity of 40500 N and 6 operates on the following work cycles.

- (i) radial load of 5000 N at 500 rpm for 25% of the time;
  - (ii) radial load of 10000 N at 700 rpm for 50% of the time; and
  - (iii) radial load of 7000 N at 400 rpm for the remaining 25% of the time.
- Calculate the expected life of the bearing in hours.

B) A single-row deep groove ball bearing is subjected to a radial force of 8 kN and a thrust force of 3 kN. The values of X and Y factors are 0.56 and 1.5 respectively. The shaft rotates at 1200 rpm. The diameter of the shaft is 75 mm and Bearing No. 6315 (C = 112000 N) is selected for this application.

- (i) Estimate the life of this bearing, with 90% reliability.
- (ii) Estimate the reliability for 20 000 h life.

Q3. Solve the following.

- A) It is required to design a pair of spur gears with  $20^\circ$  full-depth involute teeth consisting of a 20-teeth pinion meshing with a 50 teeth gear. The pinion shaft is connected to a 22.5 kW, 1450 rpm electric motor. The starting torque of the motor can be taken as 150% of the rated torque. The material for the pinion is plain carbon steel Fe 410 ( $S_{ut} = 410 \text{ N/mm}^2$ ), while the gear is made of grey cast iron FG 200 ( $S_{ut} = 200 \text{ N/mm}^2$ ). The factor of safety is 1.5.

1. Identify which of the gear elements is weaker through proper calculation.
2. Assume pitch line velocity 5 m/s & estimate the module based on beam strength.
3. Calculate important dimensions of the gears.

**Use Given data from tables for selection of various factors.**

- B) Explain following terminologies of Spur Gear, with necessary formulae.

1. Velocity ratio
2. Module of the gear
3. Circular Pitch

Q4. Solve Any Two of the following.

- A) A pair of bevel gears, with  $20^\circ$  pressure angle, consists of a 20 teeth pinion meshing with a 30 teeth gear. The module is 4 mm, while the face width is 20 mm. The material for the pinion and gear is steel 50C4 ( $S_{ut} = 750 \text{ N/mm}^2$ ). The gear teeth are lapped and ground (Class-3) and the surface hardness is 400 BHN. The pinion rotates at 500 rpm and receives 2.5 kW power from the electric motor. The starting torque of the motor is 150% of the rated torque. Effective load on gear tooth is 3443.74 N.

1. Calculate no. of virtual teeth on pinion
2. Calculate Lewis form factor.
3. Calculate PCD of pinion and gear
4. Calculate Cone Distance
5. Calculate Beam strength
6. Calculate Factor of safety against bending failure.

**Refer Data Table for value of Lewis form factor.**

- B) 1 kW power at 720 rpm is supplied to the worm shaft. The number of starts for 6 threads of the worm is four with a 50 mm pitch-circle diameter. The worm wheel has 30 teeth with 5 mm module. The normal pressure angle is  $20^\circ$ .

Calculate the efficiency of the worm gear drive and the power lost in friction if coefficient of friction is 0.035.

- C) Explain following terminologies of worm gear with mathematical equations 6 (wherever necessary)

1. Axial Pitch
2. Lead
3. Lead Angle

Q5. Solve Any Two of the following.

- A) A chain drive is used in a special purpose vehicle. The vehicle is run by a 30kW 6 rotary engine. There is a separate mechanical drive from the engine shaft to the intermediate shaft. The driving sprocket is fixed to this intermediate shaft. The efficiency of the drive between the engine and the intermediate shafts is 90%. The driving sprocket has 17 teeth and it rotates at 300 rpm. The driven sprocket rotates at 100 rpm.

Assume moderate shock conditions and select a suitable four-strand chain for this drive.

**Use Given data from tables for selection of various factors.**

- B) Compare Belt Drives and Chain Drives.

- C) It is required to select a flat belt drive for a compressor running at 720 rpm, which is driven by a 25 kW, 1440 rpm motor. Space is available for a centre distance of 3 m. The belt is open-type. Use belt velocity as 18 m/s.

**Use Given data from tables for selection of various factors.**

Q6. Solve Any Two of the following.

- A) Differentiate between brakes and clutches. 6
- B) A solid cast iron disk, 1 m in diameter and 0.2 m thick, is used as a flywheel. It is rotating at 350 rpm. It is brought to rest in 1.5 s by means of a brake. Calculate

- (i) the energy absorbed by the brake, and  
(ii) the torque capacity of the brake.

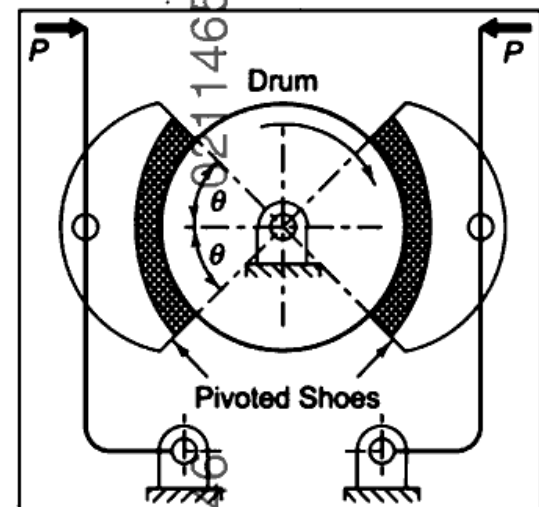
Take mass density of cast iron =  $7200 \text{ kg/m}^3$  and Radius of gyration of a solid disk

about its axis of rotation  $= \frac{d_{disc}}{\sqrt{8}}$

A pivoted double-block brake, has two shoes, which subtend an angle ( $2\theta$ ) of  $100^\circ$ . The diameter of the brake drum is 500 mm and the width of the friction lining is 100 mm. The coefficient of friction is 0.2 and the maximum intensity of pressure between the lining and the brake drum is  $0.5 \text{ N/mm}^2$ . The pivot of each shoe is located in such a manner that the moment of the frictional force on the shoe is zero.

Calculate:

- (i) the distance of the pivot from the axis of the brake drum;  
(ii) the torque capacity of each shoe; and  
(iii) the reactions at the pivot.



## Design Data

**Table 17.3** Values of Lewis Form Factor  $Y$  for 20° Full Depth Involute System

| $z$ | $Y$   | $z$ | $Y$   | $z$  | $Y$   |
|-----|-------|-----|-------|------|-------|
| 15  | 0.289 | 27  | 0.348 | 55   | 0.415 |
| 16  | 0.295 | 28  | 0.352 | 60   | 0.421 |
| 17  | 0.302 | 29  | 0.355 | 65   | 0.425 |
| 18  | 0.308 | 30  | 0.358 | 70   | 0.429 |
| 19  | 0.314 | 35  | 0.364 | 75   | 0.433 |
| 20  | 0.320 | 40  | 0.367 | 80   | 0.436 |
| 21  | 0.326 | 45  | 0.373 | 90   | 0.442 |
| 22  | 0.330 | 50  | 0.380 | 100  | 0.446 |
| 23  | 0.333 |     | 0.386 | 150  | 0.458 |
| 24  | 0.337 |     | 0.389 | 200  | 0.463 |
| 25  | 0.340 |     | 0.399 | 300  | 0.471 |
| 26  | 0.344 |     | 0.408 | Rack | 0.484 |

**Table 14.2** Power rating of simple roller chain

| Pinion speed (rpm) | Power (kW) |      |      |       |       |       |       |       |
|--------------------|------------|------|------|-------|-------|-------|-------|-------|
|                    | 06 B       | 08 A | 08 B | 10 A  | 10 B  | 12 A  | 12 B  | 16 B  |
| 50                 | 0.14       | 0.28 | 0.34 | 0.53  | 0.64  | 0.94  | 1.07  | 2.59  |
| 100                | 0.25       | 0.53 | 0.64 | 0.98  | 1.18  | 1.74  | 2.01  | 4.83  |
| 200                | 0.47       | 0.98 | 1.18 | 1.83  | 2.19  | 3.40  | 3.75  | 8.94  |
| 300                | 0.61       | 1.34 | 1.70 | 2.68  | 3.15  | 4.56  | 5.43  | 13.06 |
| 500                | 1.09       | 2.24 | 2.72 | 4.34  | 5.01  | 7.69  | 8.53  | 20.57 |
| 700                | 1.48       | 2.95 | 3.66 | 5.91  | 6.71  | 10.73 | 11.63 | 27.73 |
| 1000               | 2.03       | 3.94 | 5.09 | 8.05  | 8.97  | 14.32 | 15.65 | 34.89 |
| 1400               | 2.73       | 5.28 | 6.81 | 11.18 | 11.67 | 14.32 | 18.15 | 38.47 |
| 1800               | 3.44       | 6.98 | 8.10 | 8.05  | 13.03 | 10.44 | 19.85 | —     |
| 2000               | 3.80       | 6.26 | 8.67 | 7.16  | 13.49 | 8.50  | 20.57 | —     |

**Table 14.3** Service factor ( $K_f$ )

| Type of driven load                                                                     | Type of input power            |                |                                 |
|-----------------------------------------------------------------------------------------|--------------------------------|----------------|---------------------------------|
|                                                                                         | IC engine with hydraulic drive | Electric motor | IC engine with mechanical drive |
| (i) Smooth: agitator, fan, light conveyor                                               | 1.0                            | 1.0            | 1.2                             |
| (ii) Moderate shock: machine tools, crane, heavy conveyor, food mixer, grinder          | 1.2                            | 1.3            | 1.4                             |
| (iii) Heavy shock: punch press, hammer mill, reciprocating conveyor, rolling mill drive | 1.4                            | 1.4            | 1.7                             |

**Table 14.4** Multiple strand factor ( $K_s$ )

| Number of strands | $K_s$ |
|-------------------|-------|
| 1                 | 1.0   |
| 2                 | 1.7   |
| 3                 | 2.5   |
| 4                 | 3.3   |
| 5                 | 3.9   |
| 6                 | 4.6   |

## Design Data for Belts Selection

**Table 13.1** Load correction factor ( $F_d$ )

| Type of load                                                                                                                | $F_d$ |
|-----------------------------------------------------------------------------------------------------------------------------|-------|
| (i) Normal load                                                                                                             | 1.0   |
| (ii) Steady load, e.g., centrifugal pumps, fans, light machine tools, conveyors                                             | 1.2   |
| (iii) Intermittent load, e.g., heavy duty fans, blowers, compressors, reciprocating pumps, line shafts, heavy-duty machines | 1.3   |
| (iv) Shock load, e.g., vacuum pumps, rolling mills, hammers, grinders                                                       | 1.5   |

**Table 13.2** Arc of contact factor ( $F_d$ )

| $\alpha_s$<br>(Deg.) | 120  | 130  | 140  | 150  | 160  | 170  | 180  | 190  | 200  |
|----------------------|------|------|------|------|------|------|------|------|------|
| $F_d$                | 1.33 | 1.26 | 1.19 | 1.13 | 1.08 | 1.04 | 1.00 | 0.97 | 0.94 |

|          |                                |
|----------|--------------------------------|
| HI-SPEED | 0.0118 kW per mm width per ply |
| FORT     | 0.0147 kW per mm width per ply |

### Standard Belt Widths

|       |     |     |     |     |     |    |     |     |     |     |
|-------|-----|-----|-----|-----|-----|----|-----|-----|-----|-----|
| 3-Ply | 25  | 40  | 50  | 63  | 76  |    |     |     |     |     |
| 4-Ply | 40  | 44  | 50  | 56  | 76  | 90 | 100 | 112 | 125 | 152 |
| 5-Ply | 76  | 100 | 112 | 125 | 152 |    |     |     |     |     |
| 6-Ply | 112 | 125 | 152 | 180 | 200 |    |     |     |     |     |

Preferred Pulley diameters (mm) of CI & MS pulleys:

100, 112, 125, 140, 160, 180, 200, 224, 250, 280, 315, 355, 400, 450, 500, 560, 630, 710, 800, 900 and 1000

\*\*\* End \*\*\*